



EAST WEST UNIVERSITY

Multi-Vendor E-Commerce Marketplace Management System

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Submitted By:

Name	Student ID
Rifat Hossain Fahim	2024-1-60-054
Mehedi Hasan Bappi	2024-1-60-086
S.M. Jeshan Mahmud	2024-1-60-093

Submitted To

Md Sabbir Hossain
CSE Department
East West University

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Project Introduction:

The Multi Vendor E-Commerce Marketplace Management System is a web-based platform that allows multiple vendors to sell their products through a single online marketplace. Customers can browse products, place orders, make payments, and track their orders. Vendors can manage their products, inventory, and orders, while administrators manage users, vendors, products, orders, and the overall marketplace. The system aims to provide a centralized, efficient, and user-friendly e-commerce environment.

Project Stakeholder:

The main stakeholders of the system are **Customers, Vendors, Administrators, Delivery Personnel, Payment Gateway Services and System Developer / Maintenance Team**. Customers purchase products from the marketplace, vendors manage and sell their products, administrators control and monitor the entire system, delivery personnel handle order delivery, and the payment gateway processes online payments. Each stakeholder interacts with the system according to their specific responsibilities.

Stakeholder	Role
Customer	Registers on the platform, browses products, places orders, makes payments, and tracks orders.
Vendor	Adds and manages products, updates inventory, receives orders, and manages sales.
Administrator	Manages users, vendors, products, orders, payments, and overall system activities.
Delivery Personnel	Receives assigned orders, manages deliveries, and updates delivery status.
Payment Gateway	Processes and verifies online payment transactions securely.
System Developer / Maintenance Team	Develops, maintains, updates, and fixes issues in the system.

Requirements Analysis:

The requirements for the **Multi-Vendor E-Commerce Marketplace Management System** were gathered through a survey conducted with **seven business owners** who sell different types of products, including electronics, fashion items, groceries, perfumes, handicrafts, and mobile accessories.

The survey was conducted to understand how vendors currently manage their businesses, the challenges they face, and the features they expect from an e-commerce system. The collected responses were analyzed to identify the system requirements.

Survey Findings:

The survey revealed the following key findings:

- Most vendors currently manage their orders through **Facebook Messenger, WhatsApp, websites, or manually**, showing the need for a centralized order management system.
- Almost all vendors want to accept **multiple payment methods**, including Cash on Delivery (COD), bKash, Nagad, bank transfer, cards, and other online payment options.
- Vendors currently manage inventory using **applications, spreadsheets, managers, or manual methods**, indicating the need for an automated inventory management system.
- Most vendors expect **instant notifications** whenever a new order is placed.
- Customers usually contact vendors through **Facebook, WhatsApp, phone calls, help centers, or helplines**. Therefore, the system should provide multiple communication options.
- Most businesses rely on **third party courier services** such as Pathao and Steadfast for order delivery and tracking.
- Only a few vendors can generate sales reports, while most cannot. This indicates a strong need for automated report generation.
- Product return and refund policies differ among vendors, suggesting that the system should allow vendors to define their own return and refund policies.

Functional Requirement:

Stakeholder	Goal	Functional Requirement	Non-Functional Requirement
Administrator	Manage the marketplace efficiently and monitor all business activities.	The system shall allow administrators to approve/reject vendor registrations, manage vendors, customers, products, orders, and generate sales reports.	The system shall provide secure access, maintain 99.9% uptime, and generate reports within 5 seconds.
Vendor/Seller	Increase sales and manage products, orders, and inventory efficiently.	The system shall allow vendors to add, edit, and delete products, manage inventory, receive instant order notifications, generate invoices, and configure return/refund policies.	The vendor dashboard shall load within 3 seconds, support large product inventories, and provide reliable real-time notifications.
Customer/Buyer	Find products easily, purchase securely, and track orders conveniently.	The system shall allow customers to search and filter products, place orders, make payments using multiple methods (COD, bKash, Nagad, Card, Bank), track orders, and contact vendors.	The system shall provide a user-friendly interface, display search results within 2 seconds, and ensure secure payment processing.
Delivery Personnel / Courier Service	Deliver products on time and update delivery progress.	The system shall display assigned deliveries, update delivery status, and integrate with courier services (e.g., Pathao, Steadfast).	The system shall synchronize delivery status updates within 5 seconds and ensure accurate tracking information
Payment Gateway	Process customer payments securely and accurately.	The system shall verify payments, record transactions, support multiple payment methods, and process refunds when required.	The payment system shall ensure encrypted transactions, high reliability, and continuous availability.
System Developer / Maintenance Team	Maintain system performance, reliability, and scalability.	The system shall maintain activity logs, perform automatic backups, monitor system errors, and support software updates.	The system shall support at least 100,000 concurrent users, allow easy maintenance, and ensure data backup and recovery.

Usage Scenario:

A Usage Scenario describes how a user interacts with the Multi Vendor E-Commerce Marketplace Management System to complete a specific task. It explains the sequence of actions from the user's perspective.

For example, a customer logs into the system, searches for a product, views its details, adds it to the shopping cart, places an order, selects a payment method, and completes the payment. The vendor then receives the order and prepares it for delivery, while the delivery personnel delivers the product and updates the delivery status. This scenario demonstrates the overall interaction between different stakeholders during an online purchase.

1) Account Management

1.1 Create Account

User Perspective

A user (Customer or Vendor) must create an account before accessing the system. The user provides:

- Full Name
- Email Address
- Phone Number
- Password
- Address
- Account Type (Customer/Vendor)

System Perspective

The system validates the information, creates the account, and sends a verification email or OTP.

1.2 Login

User Perspective

The user enters their email and password to access the marketplace.

System Perspective

The system verifies the credentials and redirects the user to the appropriate dashboard.

2) Vendor Management

2.1 Register as Vendor

User Perspective

A seller submits business information including:

- Business Name
- Business Address
- Contact Number
- Trade License (Optional)
- Bank/Mobile Banking Details

System Perspective

The system stores the application and notifies the administrator for approval.

2.2 Manage Products

Vendor Perspective

The vendor can:

- Add Products
- Edit Products
- Delete Products
- Upload Images
- Set Price
- Manage Stock Quantity

System Perspective

The system updates the product catalog and inventory in real time.

3) Product Browsing

3.1 Search Products

Customer Perspective

Customers search products by:

- Product Name
- Category
- Brand
- Price Range

System Perspective

The system displays matching products with filtering and sorting options.

3.2 View Product Details

Customer Perspective

Customers can view:

- Product Images
- Description
- Price
- Ratings
- Reviews
- Vendor Information
- Stock Availability

System Perspective

The system retrieves product information from the database.

4) Shopping Cart Management

4.1 Add to Cart

Customer Perspective

Customers add products to their shopping cart before checkout.

System Perspective

The system stores selected items and calculates the subtotal automatically.

4.2 Update Cart

Customer Perspective

Customers can increase or decrease product quantity or remove items.

System Perspective

The cart total is updated instantly.

5) Order Management

5.1 Place Order

Customer Perspective

Customers confirm:

- Delivery Address
- Payment Method
- Order Summary

System Perspective

The system creates the order and sends notifications to the customer and vendor.

5.2 Track Order

Customer Perspective

Customers can monitor order status such as:

- Pending
- Confirmed
- Processing
- Shipped
- Delivered
- Cancelled

System Perspective

The system updates order status based on vendor/admin actions.

6) Payment Management

6.1 Make Payment

Customer Perspective

Customers can pay using:

- Cash on Delivery
- bKash
- Nagad
- Debit/Credit Card
- Online Banking

System Perspective

The system verifies payment and updates payment status.

7) Inventory Management

7.1 Manage Inventory

Vendor Perspective

The vendor monitors stock levels and updates inventory whenever products are sold or restocked.

System Perspective

The system automatically reduces stock after each successful order.

8) Review & Rating

8.1 Submit Review

Customer Perspective

After receiving the product, customers can rate and review it.

System Perspective

The system stores reviews and updates the product rating.

9) Notification Management

9.1 Receive Notifications

User Perspective

Customers and vendors receive notifications for:

- New Orders
- Payment Confirmation
- Order Status Updates
- Promotional Offers

System Perspective

The system sends notifications through email, SMS, or in-app alerts.

10) Administration

10.1 Manage Marketplace

Administrator Perspective

The administrator can:

- Approve Vendors
- Manage Customers
- Manage Products
- Manage Categories
- Monitor Orders
- Verify Payments
- Generate Reports
- Suspend Fraudulent Accounts

System Perspective

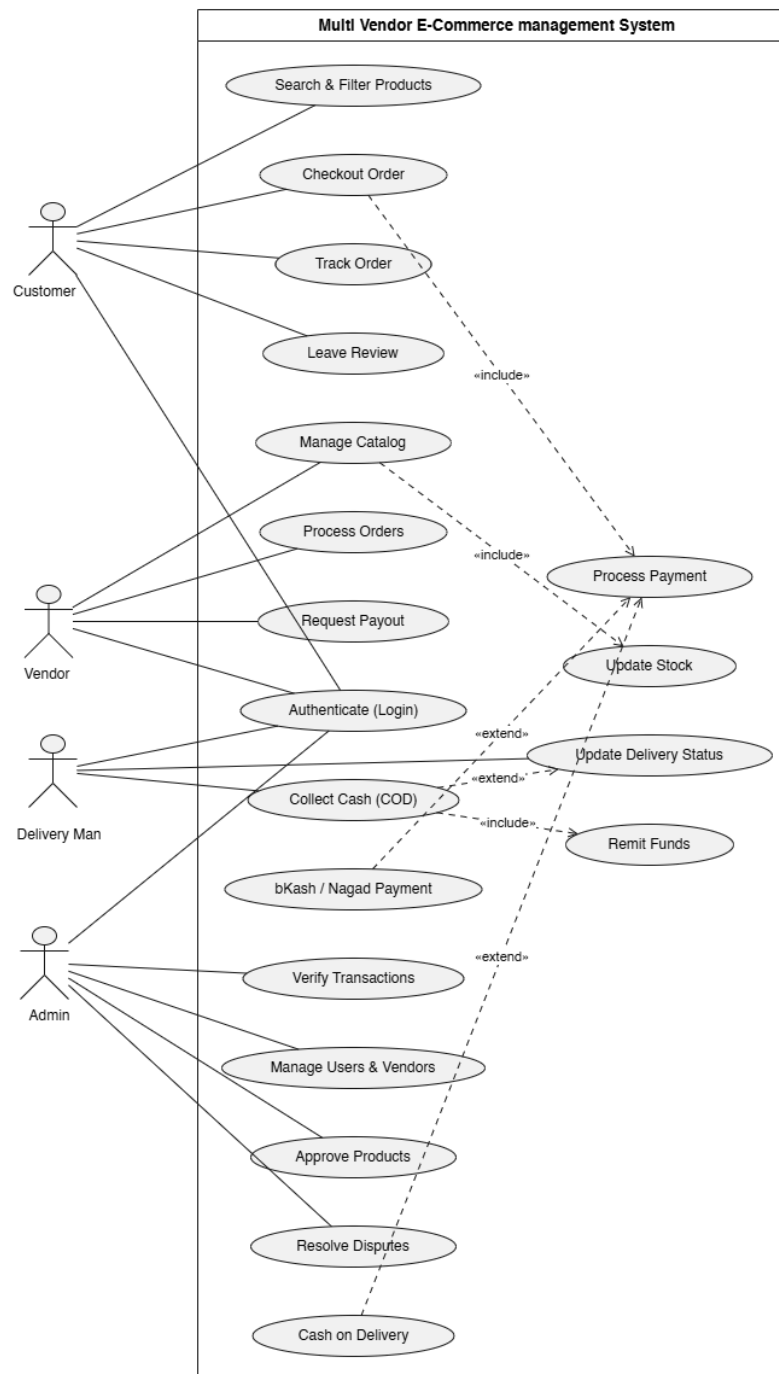
The system records all administrative activities and maintains marketplace security.

Actors in the System

- Administrator
- Vendor (Seller)
- Customer
- Delivery Personnel
- Payment Gateway

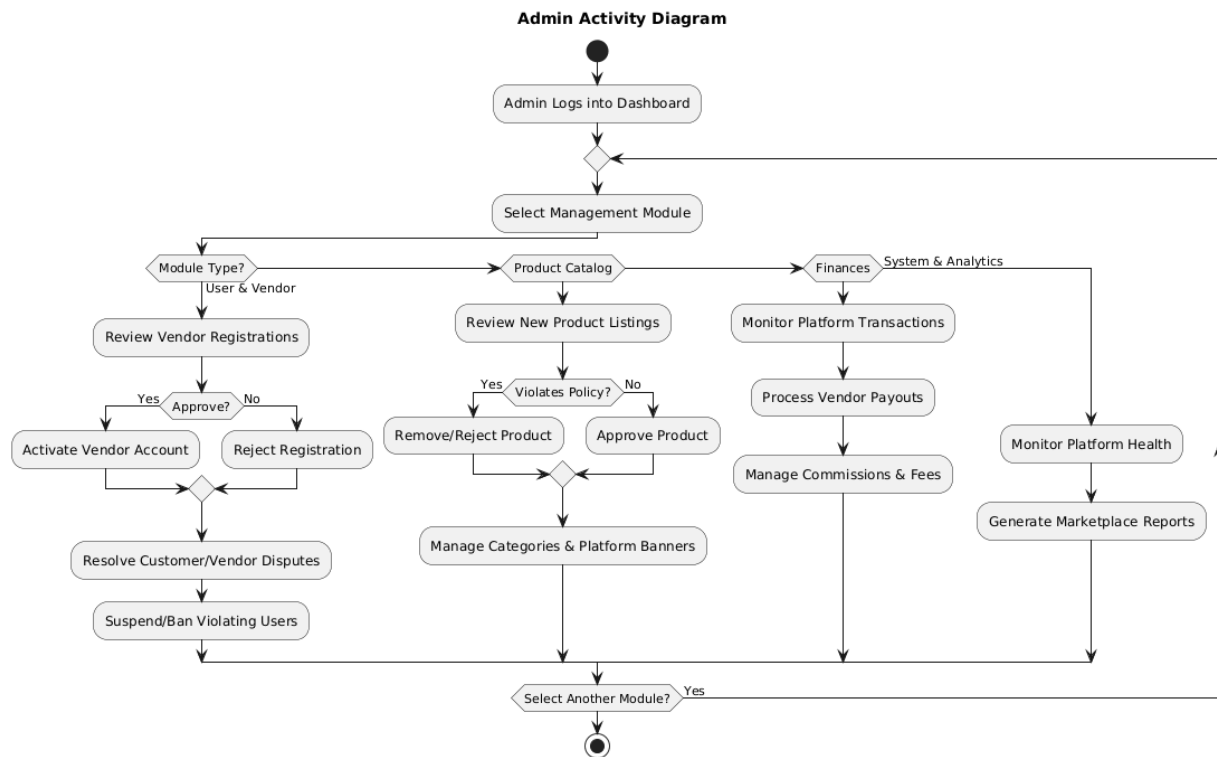
Use Case Diagram:

The use case scenario describes how different users interact with the system to complete their activities. Customers can register, log in, browse products, add products to the cart, place orders, make payments, and track orders. Vendors can manage products, inventory, and customer orders. Administrators can manage users, vendors, products, orders, payments, and system activities. Delivery personnel can view assigned orders and update delivery status.

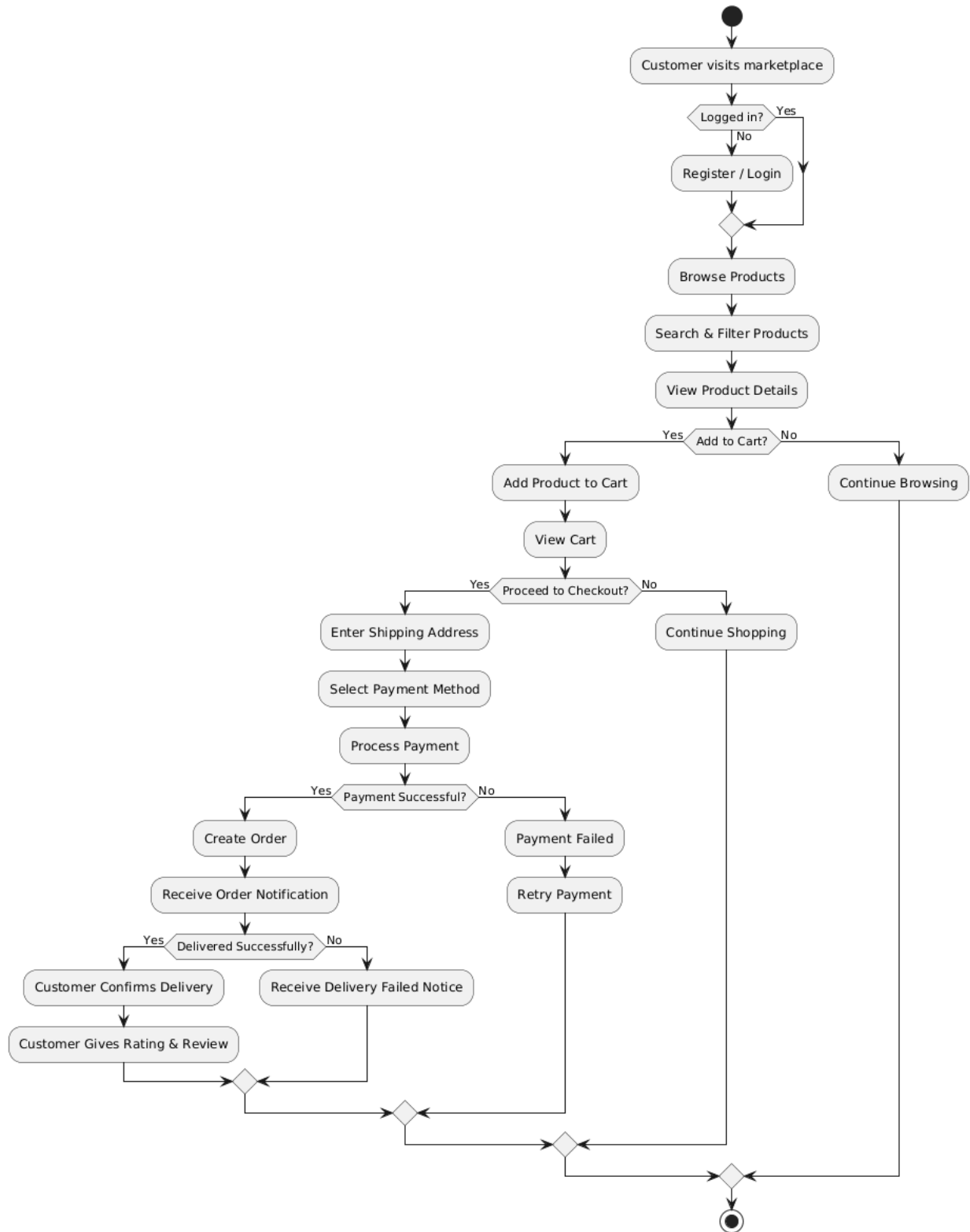


Activity Diagram:

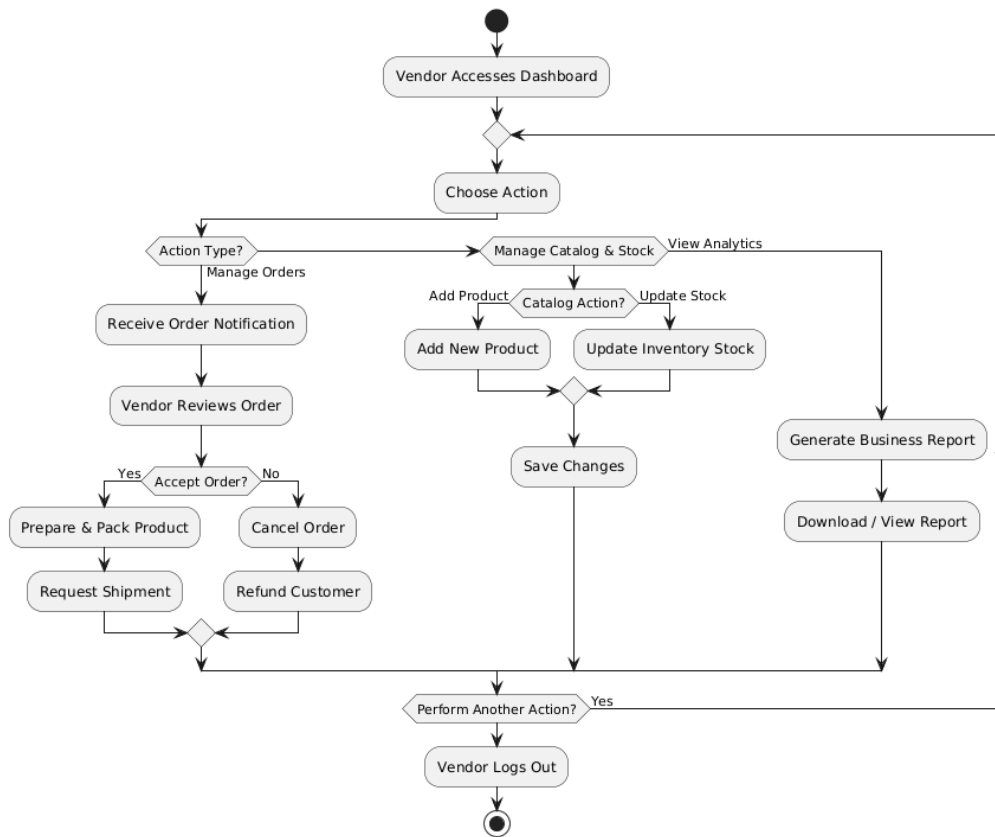
The Activity Diagram represents the workflow of important activities performed within the system. It illustrates the sequence of actions and decisions involved in processes such as customer registration, product purchasing, order placement, payment, and delivery. The diagram helps demonstrate how the system moves from one activity to another until a process is completed.



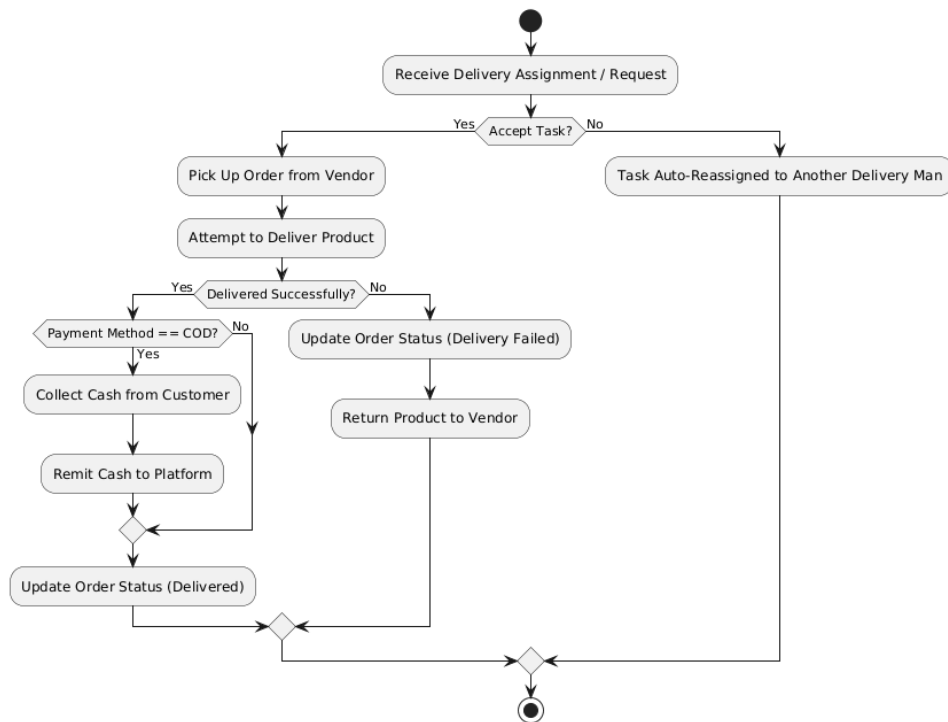
Customer Activity Diagram



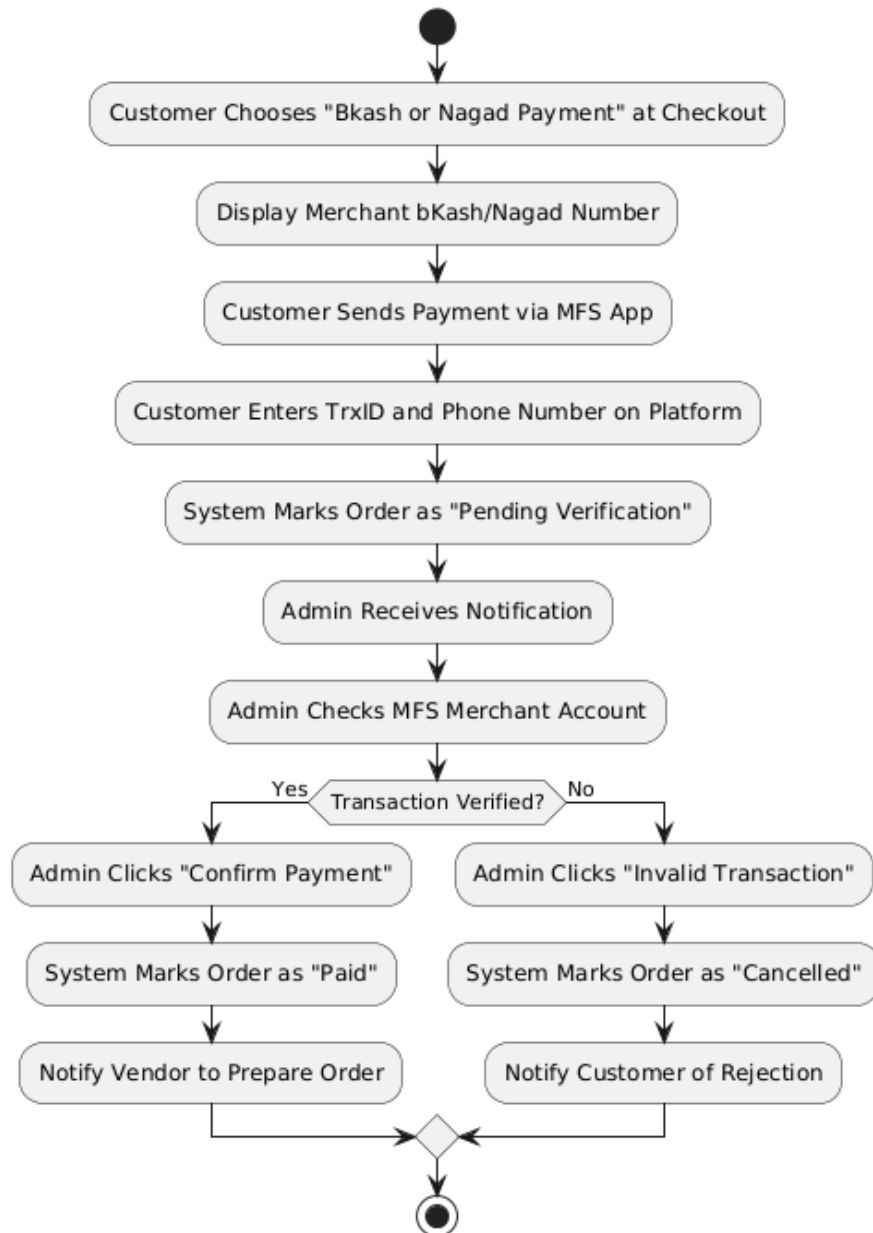
Vendor Activity Diagram



Delivery Man Activity Diagram

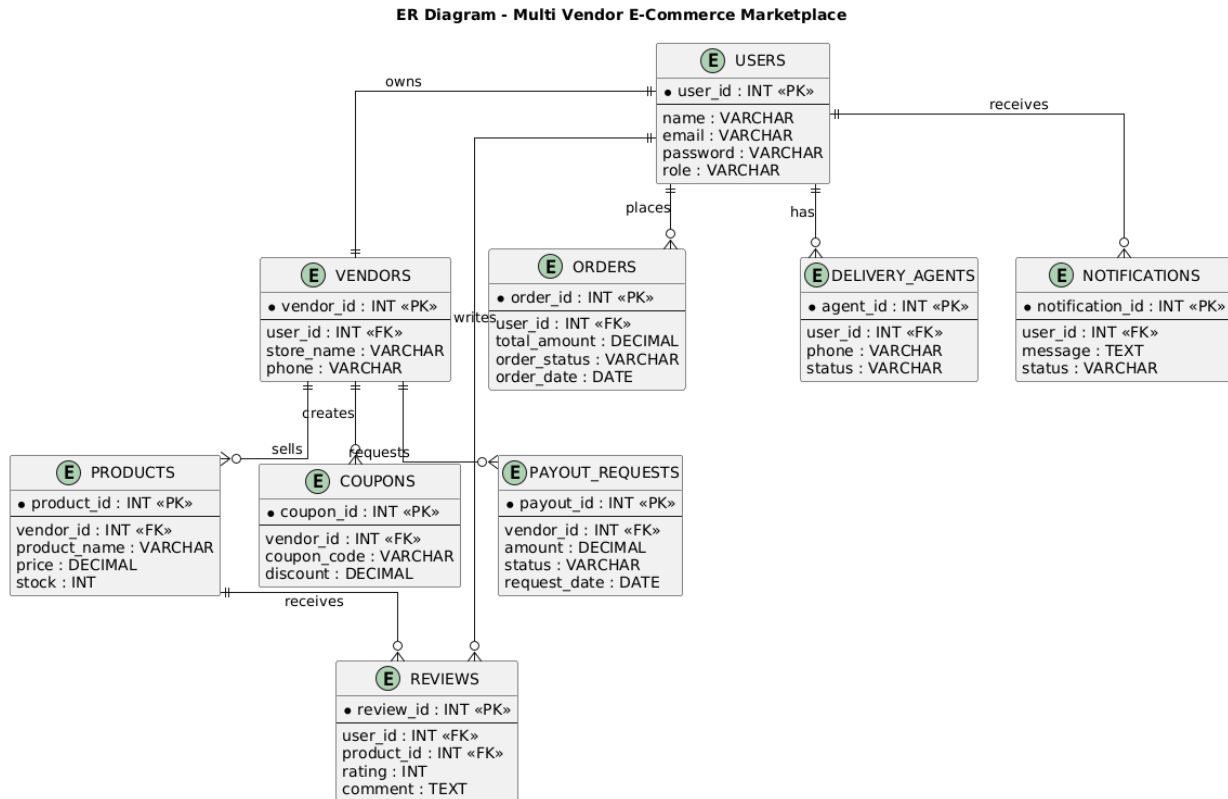


Manual Payment (bKash/Nagad) Activity Diagram



ER Diagram :

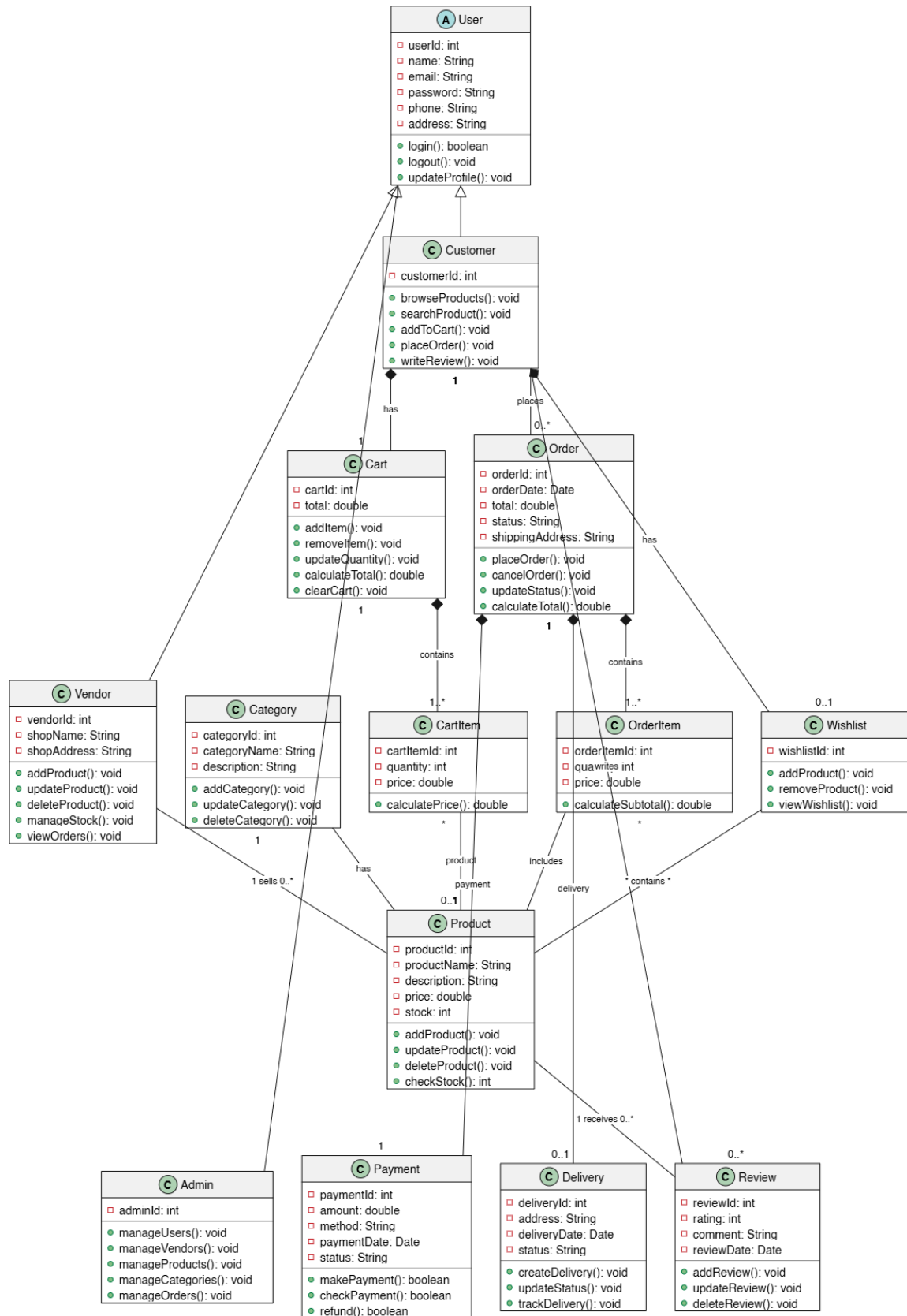
The Entity-Relationship (ER) Diagram represents the logical structure of the system's database. It identifies the major entities, their attributes, and the relationships between them. Important entities include Customer, Vendor, Product, Category, Order, Order Item, Payment, Delivery, and Admin. The diagram shows how these entities are connected and helps ensure proper database organization and data integrity.



Class Diagram:

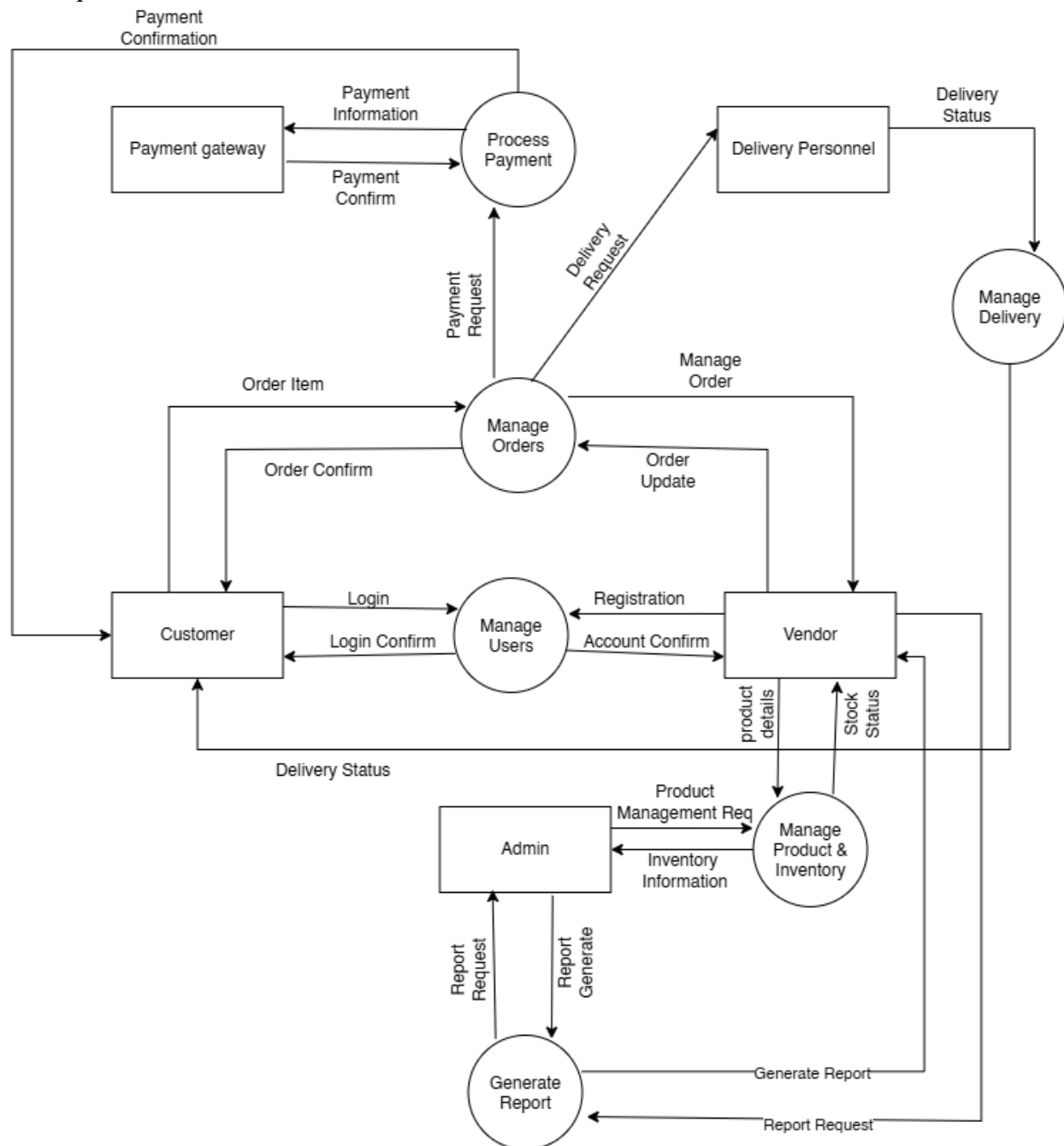
The Class Diagram represents the static structure of the system using classes, attributes, methods, and relationships. It describes the main objects involved in the application, such as Customer, Vendor, Admin, Product, Order, Payment, and Delivery. The diagram also shows relationships such as association, inheritance, and dependency between different classes.

Multi Vendor E-Commerce Marketplace Management System



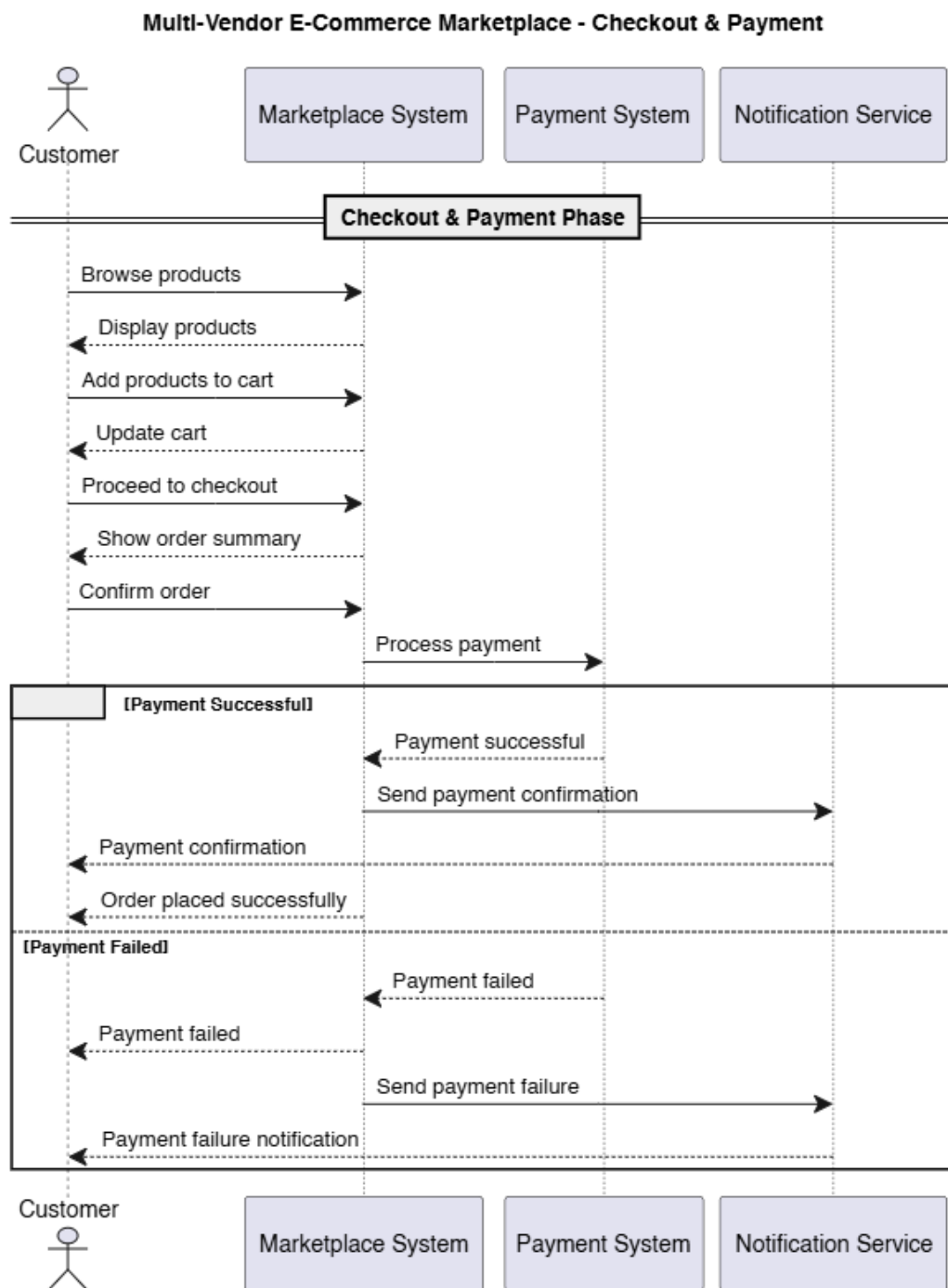
Data Flow Diagram (DFD):

The Data Flow Diagram illustrates how data moves through the system. It shows the interaction between external entities, system processes, and data stores. Major data flows include customer information, product information, order details, payment information, and delivery information. The DFD helps explain how data is received, processed, stored, and transferred within the marketplace.

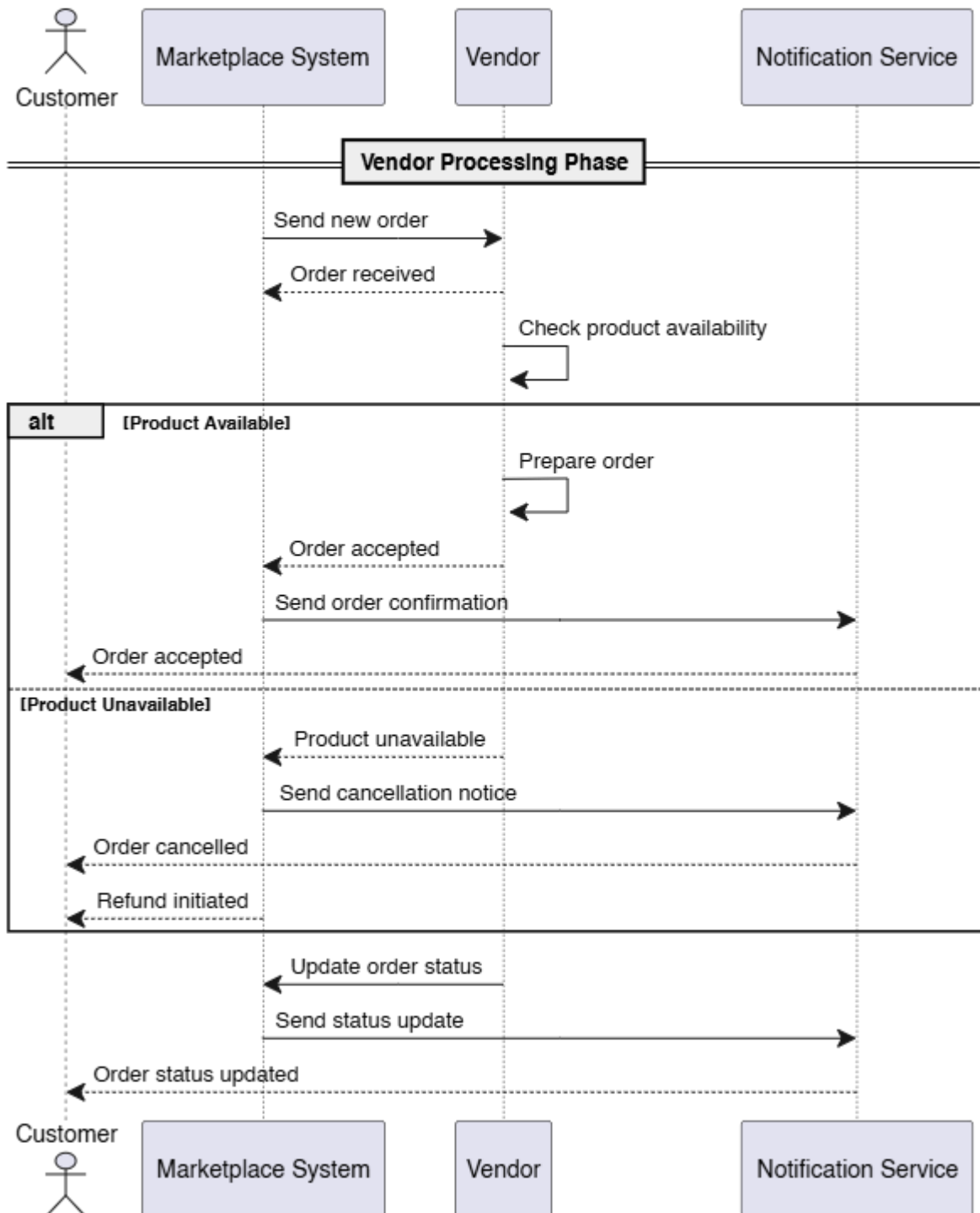


Sequence Diagram:

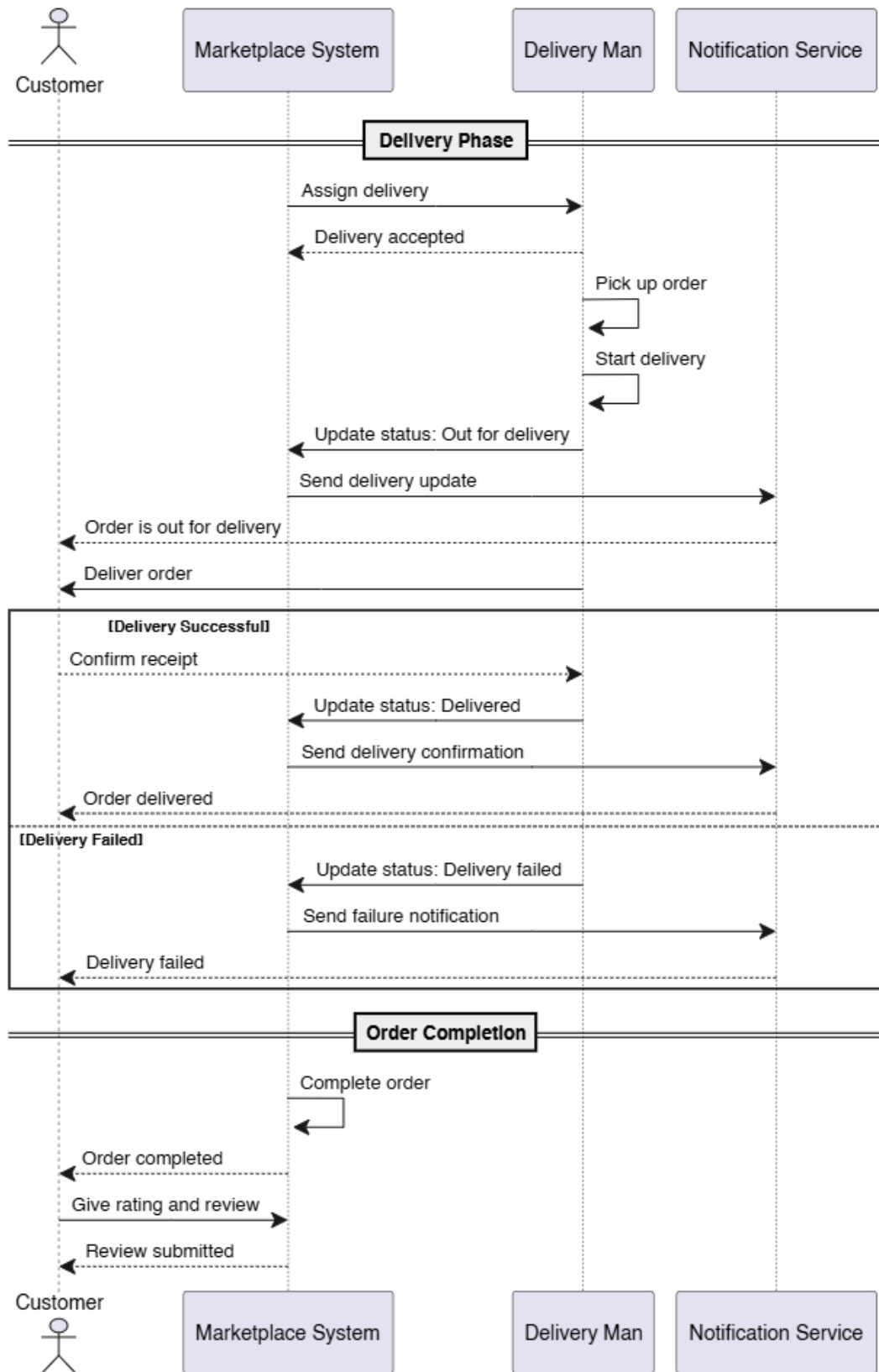
The Sequence Diagram illustrates the interaction between system components in a chronological order. It shows how different actors and system objects communicate with each other during a particular process. For example, during an order placement process, the Customer interacts with the system, product and order information is processed, payment is verified, and the order is forwarded for delivery.



Multi-Vendor E-Commerce Marketplace - Vendor Processing

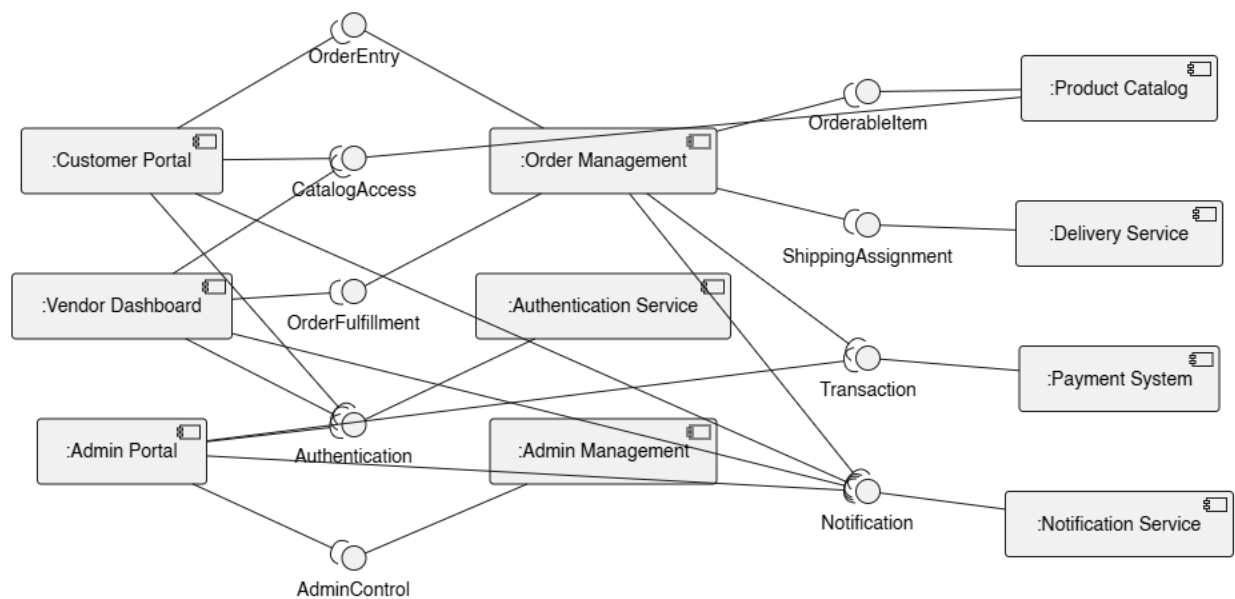


Multi-Vendor E-Commerce Marketplace - Delivery & Order Completion



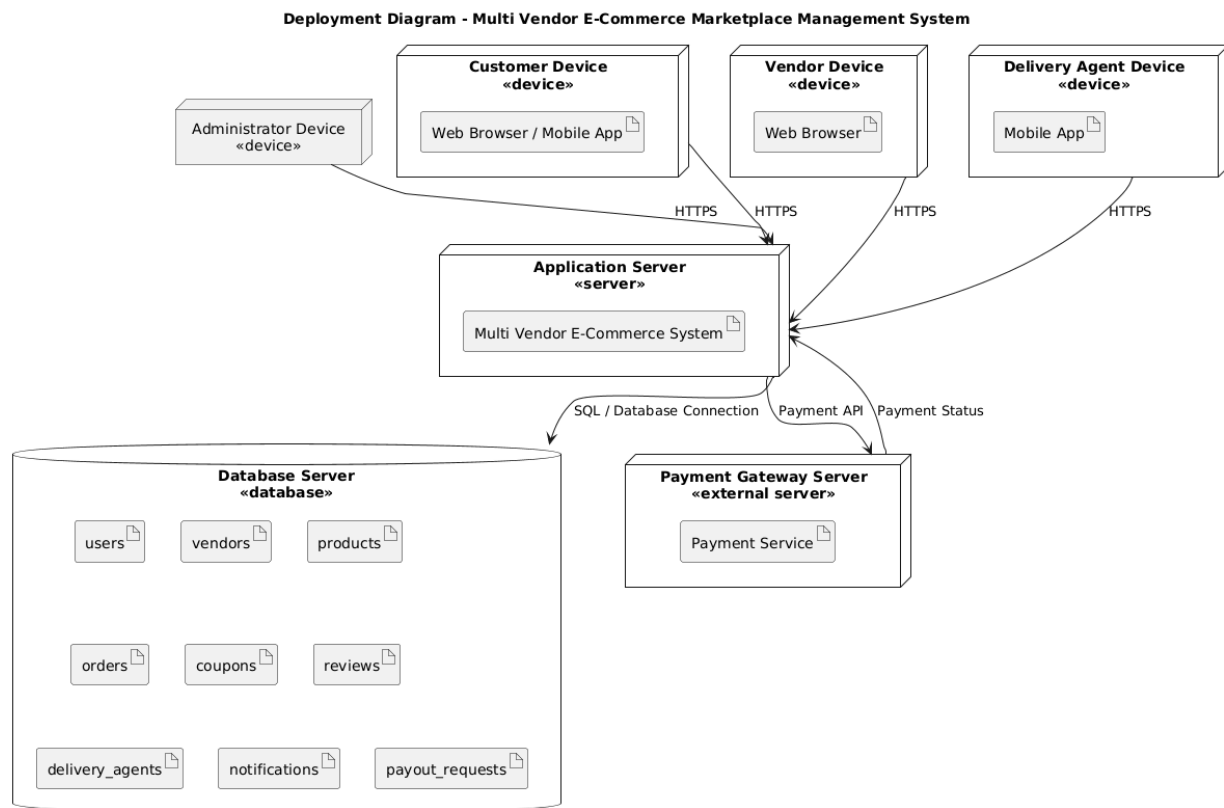
Component Diagram:

The Component Diagram represents the major software components of the system and the relationships between them. The system can be divided into components such as User Management, Product Management, Order Management, Payment Management, Vendor Management, Delivery Management, Database, and Authentication. It shows how these components communicate to provide the complete functionality of the marketplace.



Deployment Diagram:

The Deployment Diagram describes the physical deployment architecture of the system. It shows the hardware and software environments where different parts of the application are executed. The main nodes may include the user's device, web browser, application server, database server, and external payment gateway. The diagram demonstrates how these nodes communicate over the network.



Conclusion:

The Multi-Vendor E-Commerce Marketplace Management System provides a centralized platform for customers, vendors, administrators, and delivery personnel. The system simplifies product management, online purchasing, payment processing, order management, and delivery tracking. The different UML and system diagrams provide a clear understanding of the system's structure, behavior, data flow, and deployment architecture. Overall, the proposed system can improve the efficiency, scalability, and convenience of managing a multi-vendor online marketplace.

Appendix:

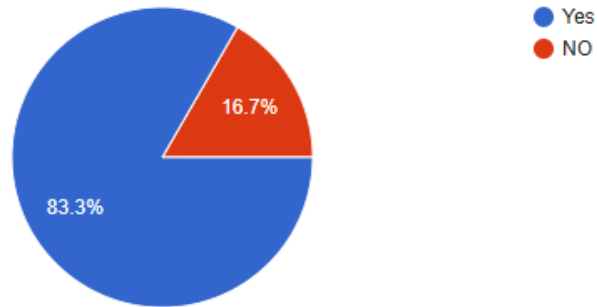
Survey and Interview Documentation:



Do you use any app or website to generate invoices?

[Copy chart](#)

6 responses



How do you currently manage your orders?

7 responses

✦ Summarize responses

Facebook, Website
Manually
Othoba.com , Offline
E-commerce and offline outlet
By Facebook messenger and ads
Online and Network
Through facebook /WhatsApp

Business Shop Name

7 responses

Vision
Novalo
Walkar
Daily Shopping
Easy Gadgets
Scentologic luxury perfume
Tanvent

What Product Do You Sell?

7 responses

✦ Summarize responses

Electronics TV , Fridge AC etc
Jersey and Fashion item
Shoe,sangle,bag
Groceries
Phone gadgets and accessories
Inspired Perfume
Handicrafts

Through online
From computer and payment method
cod
Online bank card
cod , online
Online Mkt or Offline
cash , card , cash

✦ Summarize responses

7 responses

How should customers pay for products?

How do you want to manage your product inventory?

6 responses

✦ Summarize responses

App
Sheet
...
I have a manager for that
Import and Authentic party
Using application

How should the system notify you about new orders?

5 responses

✦ Summarize responses

....

I don't have something like that

Instantly

My Online Pages,Whatsapp,Instagram and Social media Platform

Though call or sms

How do customers track their orders?

5 responses

✦ Summarize responses

3rd party Delivery (steadfast, pathao)

No way

E-commerce and Delivery platform

As I use Pathao for my deliveries

From online courier services

How should customers contact you if they have questions?

7 responses

Help center

Via Mobile Number

Notification

Helpline

They message me for asking questions

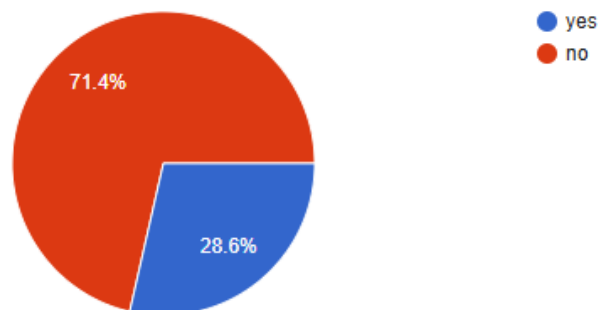
My Facebook page and Whatsapp

Using application

Can you generate daily, weekly, monthly sales reports?

7 responses

 Copy chart



which payment methods are accepted?

7 responses

✦ Summarize responses

Bkash,nagad, Bank ,COD

bKash Nagad Bank Transfer

Bkash,nagad,Bank

BKASH,NAGAD UPAY,CARD

Only by cash

Bkash

Cash or online

How are product returns and refunds handled?

3 responses

✦ Summarize responses

From Pathao Courier

Only exchanges and Packaging cost

Not decided yet

Github:

<https://github.com/rifat104/Multi-Vendor-E-Commerce-Marketplace-Management-System>

Live Project Link:

<https://multi-vendor-e-commerce-marketplace-seven.vercel.app/>